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Humza Khan

Product Engineer · AI safety & security for agents · LLM evaluation · Learning to fine-tune

SUMMARY

Self-taught product engineer who started programming by running multiplayer game servers as a teenager; a decade of experience across financial data, mobile security, and applied AI. Currently a founding engineer at a multimodal AI startup, focused on evaluation and safeguards. Moving into empirical AI safety research; available full-time January–April 2027.

RESEARCH INTERESTS

01. **AI safety & security for agents.** Adversarial robustness and AI control: keeping agents controllable under hostile inputs, and measuring how often defenses hold.
02. **Specialised local models.** Fine-tuning small open models for narrow, high-stakes tasks, evaluated against frontier baselines.
03. **Autonomous workflows with provenance.** Agent outputs a human can verify and replay, from source to claim.

EDUCATION

Self-taught; left university to build full-time. AWS Certified Solutions Architect – Associate · Y Combinator Startup School '18.

EXPERIENCE

Founding Engineer · Stealth startup (spatial intelligence, multimodal AI) — Apr 2026 - Present

- Lead engineering end to end for a multimodal LLM pipeline (video, audio, documents) producing spatially grounded findings; built primarily in Python (Temporal, pydantic, OpenCV).
- Ran a pilot eval of a video-language model against a human-labelled reference: a prompt matrix with 5 repeated runs and consensus-of-3 raised the share of labelled issues caught from **56%** to **92%**. Showed the baseline was stably wrong, not noisy, and traced a false positive repeated in 4 of 5 runs.
- Kept the model advisory where its output affects people's money: findings lacking a timestamp, source capture, or lease basis are dropped, unsupported dollar estimates are suppressed, injected instructions in model output are filtered, and weak evidence goes to a human who makes the final call.
- Isolated each customer's homes and footage: fail-closed tenant scoping at the data layer, short-lived scoped storage URLs, revocable share links that never expose raw video, and a hash-chained, append-only audit log.

Senior Software Engineer · Integral (AI accounting platform) — Apr 2024 - Sep 2025

- Owned the ingestion pipeline as the platform grew from crypto to small-business accounting: built the source integrations, deduplicated transactions across overlapping feeds to raise data quality, and used AI heavily to build the pipelines (FastAPI, SQLAlchemy).

Senior Software Engineer · Lookout (mobile security) — Aug 2022 - Feb 2024

- Redesigned the ingestion service feeding the mobile security graph for **~12M devices** (**$60K/yr** saved, infrastructure spend down **30%**) and cut API latency **55%** through request batching and DB partitioning.

Full Stack Engineer · Revelate (financial data) — Mar 2019 - Jul 2022

- Built and scaled a real-time trade-analytics engine processing **200–400M messages/day** across NYSE, NASDAQ, and TSX (Java, Kafka, Spark, Airflow); cut metrics delivery from 30 to 12 minutes.

Principal Consultant · Self-employed — May 2016 - Feb 2019

- Took SaaS, telecom, and fintech products from ambiguous requirements to shipped MVPs, owning architecture end to end. Earlier: full-stack roles, 2015–2018.

LATEST PROJECTS

Fine-Tuned Statement Extraction `Python · in progress` · Supervised fine-tuning (LoRA) of small open models on bilingual bank statements vs. Qwen3.5-4B and NuExtract3 baselines; synthetic ground truth; sandboxed pipeline that treats PDF text as untrusted.

Provenant `Go` · `provenant.sh` · Provenance layer for agent outputs: fail-closed reads, schema-first validation, deterministic replay.

Fauxbooks `Go` · `fauxbooks.xyz` · Reconcilable synthetic accounting datasets, byte-identical across platforms, for testing financial pipelines and models.

TOOLKIT

Python, TypeScript, Go · Claude Code (Opus, Sonnet), Codex · Promptfoo, DeepEval, LLM-as-judge · Qwen, NuExtract · Temporal, Postgres, Kafka, AWS